

LAVA: DECODING-FREE LATENT VERIFICATION FOR EFFICIENT TEXT-TO-AUDIO INFERENCE-TIME SCALING

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ABSTRACT

Inference-time scaling improves diffusion without retraining: search algorithms sample multiple noise particles and use a verifier during denoising to steer sampling toward high-reward outputs. Standard methods decode every candidate and score it with an external verifier, making repeated evaluation the main computational bottleneck. We introduce a latent-space verifier: a lightweight head attached to an intermediate layer of the frozen generator that predicts reward embeddings from noisy latents, eliminating decoding and enabling efficient per-step scoring. Across Best-of- N , Particle Filtering, and Rollover Budget Forcing search on the text-to-audio model TangoFlux, our verifier achieves up to $10.7\times$ speedup while matching the Fréchet distance of decode-based CLAP rewards. Starting from the original TangoFlux baseline without inference-time scaling 72.58, our approach reduces OpenL3-FD on AudioCaps to 64.62 by reinvesting the compute saved from latent-space verification into additional particles. Our scoring head uses 4.8M parameters, $14\times$ fewer than the 68.6M-parameter CLAP audio encoder.

Index Terms— Text to audio generation, Diffusion Modeling, Inference Time Scaling

1. INTRODUCTION

Latent-space diffusion and flow-matching models are the dominant paradigm for high-fidelity generation across images, videos, and audio [1, 2, 3]. They generate samples through an iterative sampling process, with the number of sampling steps serving as the standard lever for trading compute against quality. Beyond a point, however, additional sampling steps yield diminishing returns; extra test-time compute is better invested in *inference-time scaling*. Rather than allocate additional compute to a single sampling trajectory, these methods maintain multiple candidate trajectories and use a verifier to search for promising ones, typically by scoring them with an external reward model [4]. In effect, quality is scaled not by sampling longer, but by how candidates are searched and scored.

Text-to-audio generation has similarly benefited from advances in latent generative modeling. AudioLDM [3] was among the first to apply latent diffusion to text-to-audio, using CLAP [5] embeddings to bridge audio and text conditioning. AudioLDM2 [6] unified speech, music, and general sound through a shared self-supervised representation. The Tango line [7, 8] then introduced an instruction-tuned LLM text encoder and preference optimization, while EzAudio [9] advanced efficiency and fidelity with a diffusion transformer over a learned waveform latent. Most recently, Tan-

goFlux [10] recasts generation as flow matching with CLAP-ranked preference tuning, yielding a fast, high-quality generator.

Despite the advances in audio generation, inference-time scaling for audio remains underexplored. In the image domain, a line of work scales inference by searching over the sampling process. The simplest methods, such as best-of- N (*BoN*) and zero-order, search over the initial noise, select among randomly generated candidates by scoring only the final output [4]. Finer-grained methods instead search within the denoising trajectory, scoring intermediate states through their denoised (Tweedie) estimates and steering the population toward promising candidates via particle filtering or tree search [11, 12]. Rollover Budget Forcing (*RBF*) [13] extends this to flow models, searching at every denoising step and rolling unused budget forward to harder steps. In all of these, scoring a candidate is expensive: it must first be fully denoised, decoded into a waveform and then passed through an external reward model, a cost incurred for every candidate and, in per-step methods, at every denoising step. In audio, to our knowledge only two inference-time-scaling works exist [14, 15], and both score every candidate, after having them decoded to a waveform.

A recent line of work avoids this by scoring directly in latent space. VHS [16] attaches a lightweight verifier to the hidden states of a one-step image generator, ranking candidates without decoding to pixels; since generation is single-step, it scores one near-clean latent rather than an evolving denoising trajectory. LatSearch [17] instead scores partially-denoised latents throughout video generation to resample and prune candidates, but relies on a heavyweight vision–language reward model. Most architecturally related to our work, DiNa-LRM [18] places a timestep-conditioned reward head on a frozen latent diffusion backbone, though for preference optimization of image models rather than inference-time search.

In audio, this latent-space direction is entirely unexplored. We close this gap introducing a latent verifier for text-to-audio inference-time scaling. Our design is inspired by REPA [19], which shows that a diffusion transformer’s hidden states can be aligned to a pre-trained semantic encoder through a lightweight projection; whereas REPA applies this alignment during training, we apply it to a frozen, post-trained generator to repurpose its hidden states for verification. Concretely, a small head taps an intermediate layer of a frozen TangoFlux and scores its noisy embeddings directly against the text prompt’s embedding, without decoding. Beyond skipping the decoder, this also avoids fully running the generator for every candidate: that expensive computation is spent only on the candidates the verifier keeps. Our main contribution can be summarized as follows:

- We introduce **LaVA** (**Latent Verifier for Audio** inference search), a latent-space verifier for efficient inference-time search in text-

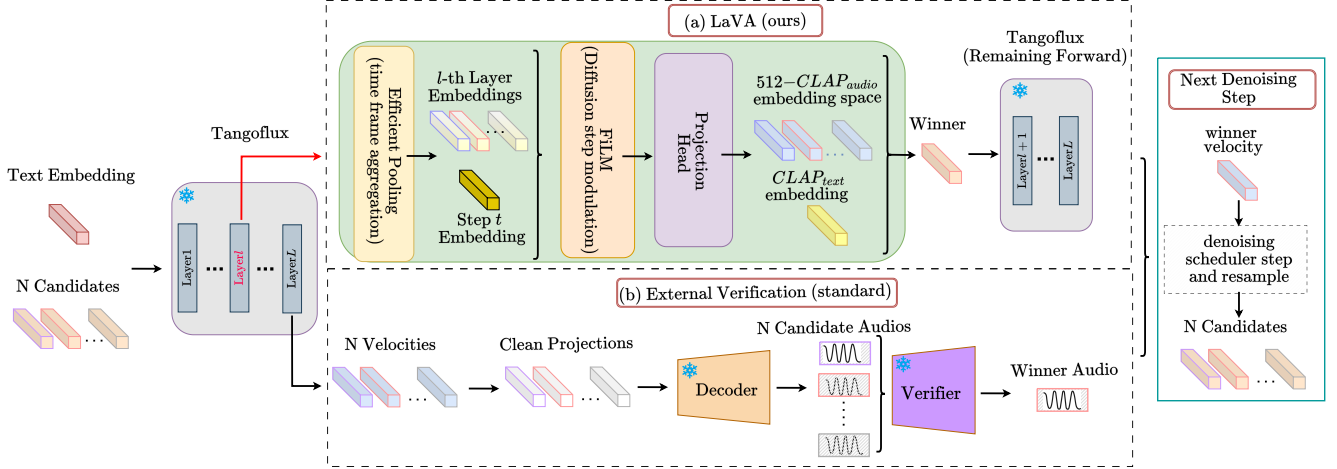


Fig. 1: LaVA search step. Unlike standard pipelines (b) that decode all N candidates for external scoring, LaVA (a) predicts CLAP rewards directly from a partial forward pass (layer ℓ). By advancing only the winning latent to the remaining layers, it avoids expensive full-trajectory evaluations and waveform decoding.

to-audio generation that replaces the decode-and-score pipeline that prior methods rely on.

- State-of-the-art per-step search methods like *RBF* evaluate many candidates at each step, running the entire transformer backbone on every candidate solely to compute its reward. Our verifier scores candidates using only the hidden representation at an intermediate layer ($\ell=6$), running the remaining layers only for the single winner candidate.
- Across *BoN*, *PF*, and *RBF* on TangoFlux, LaVA matches decode-based CLAP rewards in OpenL3-FD at up to $10.7\times$ lower wall-clock.

Code and implementation details are available on our github page, with audio comparisons and examples on our project page.

1.1. Background

We build on TangoFlux, a latent flow-matching text-to-audio model. To generate a sample, the model starts from Gaussian noise and gradually transforms it into a clean latent over T steps. At step t , with latent z_t , noise level σ_t , and prompt c , the network predicts the velocity $v_\theta(z_t, t, c)$, and one denoising step advances the latent to the next, lower noise level $\sigma_s < \sigma_t$ by an Euler update on the flow ODE,

$$z_s = z_t - (\sigma_t - \sigma_s) v_\theta(z_t, t, c). \quad (1)$$

From the same velocity, Tweedie’s formula [20] gives a one-step estimate of the clean latent,

$$\hat{z}_0(z_t) = z_t - \sigma_t v_\theta(z_t, t, c). \quad (2)$$

The clean latent z_0 , fully denoised, or the Tweedie estimate $\hat{z}_0$, is decoded to a waveform $\tilde{x} = \mathcal{D}(z_0)$, where $\mathcal{D}$ is the decoder.

Inference-time scaling generates many candidate latents, decodes each to a waveform, and picks among them based on the score of an external verifier $r(\tilde{x}, c)$. This verifier is most commonly CLAP [5], whose two encoders $\text{CLAP}_{\text{audio}}$ and $\text{CLAP}_{\text{text}}$ embed audio and text into a shared, contrastively-trained space. Verifying N particles at denoising step t of T is expensive: N decodes plus either N forward passes for their Tweedie estimates, or $N(T-t)$ to fully denoise them, all merely to rank them.

1.2. Latent Verifier for Audio Inference Time Scaling

We exploit the strong semantic representations embedded within the frozen generator to steer inference directly in latent space. Rather than decode each candidate and score it with an external model, we predict the reward directly from the noisy latent using a lightweight head attached to the frozen generator. Given a latent z_t at step t , we run the frozen TangoFlux transformer only up to layer ℓ and read its audio hidden states h_t^ℓ ; a sequence of feature vectors over the audios’s temporal frames. A small head g_ϕ maps this sequence to a predicted $\text{CLAP}_{\text{audio}}$ embedding: it aggregates the frames into a single vector with an Efficient Probing (EP) attention pool [21], modulates the result by the diffusion step t through a FiLM layer [22], and projects it with a two-layer MLP before ℓ_2 -normalizing.

$$\hat{e} = g_\phi(h_t^\ell, t) \in \mathbb{R}^{512}, \quad \|\hat{e}\|_2 = 1. \quad (3)$$

The head is conditioned on the sampling step, allowing it to score latents at arbitrary noise levels along the sampling trajectory. Together, the frozen backbone up to layer ℓ and the scoring head g_ϕ form the latent verifier.

We train only g_ϕ while keeping the backbone frozen. For each training audio clip x , we first obtain its clean latent representation z_0 by encoding the waveform with TangoFlux’s pre-trained audio encoder E , and extract its CLAP-audio embedding $e = \text{CLAP}_{\text{audio}}(x)$. We then sample a noise level σ_t from the logit-normal schedule used during TangoFlux training, together with Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$. Finally, we construct the noisy latent z_t as:

$$z_t = (1 - \sigma_t) z_0 + \sigma_t \epsilon. \quad (4)$$

and minimize the cosine distance

$$\mathcal{L}(\phi) = 1 - \cos(g_\phi(h_t^\ell, t), e). \quad (5)$$

During inference, the latent verifier scores a candidate by the cosine similarity between its embedding $\hat{e}$ and the CLAP-text embedding of the prompt Eq (6). Because g_ϕ regresses the $\text{CLAP}_{\text{audio}}$ embedding of the clean sample, scoring $\hat{e}$ against $\text{CLAP}_{\text{text}}(c)$ approximates the decode-based CLAP reward without decoding.

$$s(z_t, c) = \cos(\hat{e}, \text{CLAP}_{\text{text}}(c)). \quad (6)$$

1.3. Trajectory Search Frameworks

To generate diverse candidate proposals during denoising, we integrate our verifier with two state-of-the-art trajectory-search algorithms. We utilize the stochastic (SDE) counterpart of the ODE step in Eq (1)[13], so that repeated steps yield diverse proposals.

Particle Filtering (PF) follows the Feynman-Kac steering framework [11]. It maintains N particles denoised in parallel and periodically resamples them in proportion to potentials derived from a reward. After a 10-step warmup, it evaluates particles with the verifier every 5 steps and resamples toward high-reward trajectories, duplicating promising particles and discarding low-scoring ones.

Rollover Budget Forcing (RBF) [13] performs step-wise local search. It warm-starts by drawing M initial noise samples and keeping the top N . At each denoising step, every particle generates up to B stochastic candidate next states and transitions to one that increases its verifier reward. Unused search budget rolls forward to harder downstream steps. Since decode-based evaluation requires the Tweedie estimate $\hat{z}_0$ (and thus a full model forward per candidate to obtain velocity Eq. (2)), this creates a substantial computational bottleneck. The latent verifier resolves this by scoring candidates from a single partial forward pass, deferring the expensive full velocity computation to the chosen particle. Decoupling scoring from denoising drives most of our efficiency gains. We refer the reader to Appendix D and E of [13] for further details.

2. EXPERIMENTS

2.1. Implementation Details

Dataset. We train our latent verifier on AudioCaps [23] training split, randomly selecting one caption per 10-second clip to form 47,338 training and 2,500 validation text-audio pairs, resampled from 22.05 kHz source audio to the VAE’s 44.1 kHz stereo input.

Latent verifier. For each training (audio, caption) pair, we sample the clean VAE latent $z_0 \in \mathbb{R}^{645 \times 64}$ and noise it to σ_t using TangoFlux’s logit-normal schedule (4). The resulting z_t is passed through the frozen backbone, and audio states from the first single-stream block ($\ell = 6$, after six dual-stream MMDiT blocks) yield $h_t^\ell \in \mathbb{R}^{645 \times 1024}$. The head g_ϕ uses timestep-only FiLM conditioning and comprises EP attention pooling with 32 learnable queries (1.08M), AdaLN-FiLM (2.10M), and a Linear-GELU-Linear MLP (1.57M), producing a 512-D ℓ_2 -normalized embedding. We regress the clean CLAP-audio embedding e , extracted with LAION-CLAP [24], training only g_ϕ with AdamW (lr 10^{-4} , weight decay 10^{-2} , batch size 32) for 50 epochs and selecting the lowest-validation-loss checkpoint. Training takes ~ 2 h on a single NVIDIA-A100 GPU. The resulting 4.8M-parameter head is $14\times$ smaller than the 68.6M-parameter CLAP audio encoder, and the same head is used across all experiments.

Inference Sampling. We sample from TangoFlux following the original work using 50 denoising steps and classifier-free guidance (scale 4.5), producing audios of 44.1 KHz. All candidates for a prompt share the same Flan-T5 and CLAP-text embeddings, computed once. We report per-prompt wall-clock on a single NVIDIA-A100 GPU, timing the search call with CUDA synchronization on both ends; this includes candidate generation, verification, and the final winner’s decode (identical across all methods).

2.2. Evaluation

Following TangoFlux [10], we evaluate inference-time scaling on its 886-prompt test subset across four objective metrics: OpenL3

Fréchet Distance (OpenL3-FD) and PaSST-based KL divergence (KL) for fidelity, alongside CLAP score for prompt alignment and Inception Score (IS) for diversity. We compute OpenL3-FD, KL, and CLAP via `stable-audio-metrics`, and IS using the AudioLDM toolkit [3]. Generated audio is 44.1 kHz stereo; we retain the first 10 s and downmix to mono by channel averaging. OpenL3-FD uses the `env` model with mono embeddings, 0.5 s hop and 44.1 kHz evaluation bandwidth, against the 886 AudioCaps recordings as reference. Distinct CLAP checkpoints are used for search (`larger_clap_general`) and evaluation (`630k-audio-set-fusion-best`). All methods are compared against vanilla TangoFlux without inference-time scaling, which achieves an OpenL3-FD of 72.58 ± 2.68 across runs. Notably, inference-time scaling substantially reduces this high run-to-run variation, both with full decode-based verification and with our latent verifier.

2.3. Best of N Selection at Full and Partial Denoising

We first evaluate Best-of- N selection ($N=16$). Full decode-based CLAP scoring, which denoises and decodes all candidates before evaluation, lowers OpenL3-FD from 72.58 to 66.65 (Table 1). Our latent verifier scores noisy latents directly, avoiding full generation. Candidates are denoised only partway, and only the top-ranked sample completes the trajectory to be decoded. Selecting at just 30% of the trajectory achieves fidelity comparable to full denoising (66.85 vs. 66.65) while cutting wall-clock time by $3.0\times$.

2.4. Search Within the Denoising Trajectory

The partial-denoising results in Sec. 2.3 show that LaVA can guide candidate selection before the denoising trajectory is complete. We therefore apply it to fine-grained trajectory search with *PF* and *RBF* (Sec. 1.3). We use $N=40$ for *PF* and $B=9$, $M=16$ and $N \in \{2, 20, 30\}$ for *RBF*.

Table 1 compares both algorithms with decode-based scoring and LaVA. Both substantially improve over vanilla single-trajectory generation (72.58 OpenL3-FD): *PF* reaches 66.0, while *RBF* reaches 65.1 at $N=2$ and further improves to 64.82 and **64.62** at $N=20$ and 30, respectively. These gains do not compromise the remaining metrics, with CLAP, KL, and IS remaining comparable to the decode-based verification.

LaVA achieves this fidelity at substantially lower cost. For *PF*, all N particles are advanced at every step regardless of their scores, so LaVA primarily eliminates waveform decoding, yielding a $1.7\times$ speedup. *RBF* benefits more substantially: LaVA scores each proposal after a single partial forward and performs the full forward only for the accepted particle, yielding a $10.7\times$ wall-clock and $17.5\times$ FLOP reduction. *RBF*+LaVA at $N=20$ costs 211 s/prompt and 2186 TFLOPs and reaches 64.82, while *RBF*+CLAP at $N=2$ costs more but performs worst. LaVA make larger *RBF* populations ($N=20, 30$) practical, allowing the freed computation to be reinvested into broader search.

2.5. Ablations

Choice of ℓ and architectural depth. Figure 2 ablates the choice of intermediate layer ℓ . Under the *RBF* algorithm at $N = 2$, layers $\ell = 4, 5, 6$ yield tied results (≈ 65.1), with $\ell = 3$ performing worse (65.59). As the search scales to $N = 30$, the behavior diverges significantly. Deeper layers ($\ell = 5, 6$) continue to improve, reaching 64.80 and **64.62**, while shallower layers ($\ell = 3, 4$) plateau at ≥ 65.3 . We attribute this divergence to the increasing semantic

Method	Params	Verifier	OpenL3-FD ↓	IS ↑	KL ↓	CLAP ↑	s/prompt ↓	TFLOPs ↓
Vanilla	–	–	72.58±2.68	12.56±.07	1.0245±.03	0.543±.002	~3.4	47
BoN	$N=16$	CLAP	66.65±.06	12.80±.08	0.990±.004	0.557±.000	53.2	750
BoN	$N=16, 30\%$	LaVA	66.85±.16	12.58±.06	0.993±.004	0.549±.001	17.5 ($\times 3.0$)	245 ($\times 3.1$)
PF	$N=40$	CLAP	66.22±.19	12.88±.17	0.975±.027	0.558±.001	159.4	2799
PF	$N=40$	LaVA	66.00±.14	12.82±.10	1.008±.009	0.549±.001	94.7 ($\times 1.7$)	1789 ($\times 1.6$)
RBF	$N=2, B=9$	CLAP	65.46±.42	12.79±.13	0.991±.014	0.551±.001	225.9	3860
RBF	$N=2, B=9$	LaVA	65.13±.29	12.77±.12	0.997±.024	0.549±.002	21.1 ($\times 10.7$)	221 ($\times 17.5$)
RBF	$N=20, B=9$	LaVA	64.82±.18	12.93±.21	1.012±.012	0.552±.001	211	2186
RBF	$N=30, B=9$	LaVA	64.62±.08	12.87±.19	1.003±.013	0.551±.002	308	3373

Table 1: Inference-time scaling performance on the AudioCaps 886-prompt test subset. The **Verifier** column distinguishes our latent verifier (LaVA) from standard decode-based CLAP scoring (CLAP); blue -shaded rows use LaVA. Within each search algorithm, both variants share identical hyperparameters. Green-highlighted values denote LaVA’s speedup / FLOP reduction over its matched decode-based baseline. All quality metrics are reported as mean±std over 3 random seeds; FLOPs are measured using `fvcore`. LaVA drastically reduces computational overhead, achieving up to a $10.7\times$ speedup and $17.5\times$ fewer FLOPs under *RBF* comparing to with the decode - CLAP verification.

information captured at deeper layers, which can provide a more informative basis for evaluating complex text-audio alignment. In particular, deeper single-stream blocks have received more contextual text conditioning through the preceding cross-attention layers. Consequently, deeper layers prove substantially more stable at scale: $\ell = 6$ narrows to a variance of just ± 0.08 at $N = 30$, compared to the high noise (up to ± 0.49) observed at shallower layers.

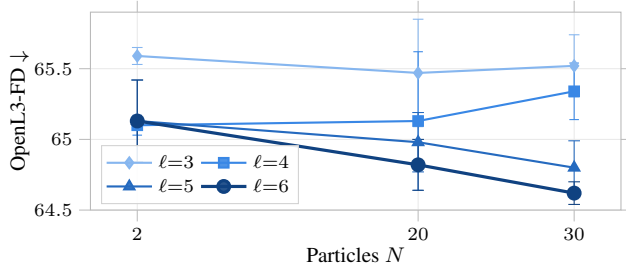


Fig. 2: Deeper layers scale. OpenL3-FD under *RBF* with increasing particle count N : deeper layers ($\ell=5, 6$) keep improving, whereas shallower layers ($\ell=3, 4$) saturate. Mean $\pm$ std are calculated across 3 seeds with $B = 9$.

Verifier objective. As reported in Table 2, our head regresses the clip’s CLAP-audio embedding under a cosine distance. Retargeting the objective directly to the caption’s CLAP-text embedding, adding Audibox perceptual-quality term [14], or substituting the regression loss with a contrastive InfoNCE objective each degrade overall OpenL3-FD.

Head Architecture Table 3 ablates the design choices within our verifier head g_ϕ . Removing the timestep FiLM modulation degrades performance to 65.77, while replacing the efficient probing (EP) attention pool with standard mean pooling drops OpenL3-FD further to 66. These results justify the inclusion of both components.

3. CONCLUSIONS

We introduced LaVA, a latent-space verifier for inference-time scaling of text-to-audio generation. By predicting the CLAP-audio rep-

Verifier objective	OpenL3-FD ↓
CLAP-audio, cosine (ours)	65.13
CLAP-text, cosine	66.16
CLAP-audio + Audibox-PQ	65.95
CLAP-audio, InfoNCE	66.48

Table 2: Verifier training objective. Evaluated using OpenL3-FD ($\downarrow$). All variants employ the *RBF* search algorithm across 3 random seeds with $N = 2$ and $B = 9$.

Head architecture	OpenL3-FD ↓
EP pool + FiLM (ours)	65.13
EP pool, no FiLM	65.77
Mean pool + FiLM	66.00

Table 3: Head architecture. Evaluated with OpenL3-FD ($\downarrow$). Removing the timestep FiLM or replacing the EP attention pool with mean pooling both degrade FD, justifying each component. All variants use *RBF* ($N=2, B=9$), 3 seeds.

resentation directly from intermediate noisy latents of a frozen TangoFlux backbone, LaVA enables candidate evaluation without waveform decoding or full-trajectory generation. This decouples verification from denoising and substantially reduces the cost of search. Across Best-of-(N), particle filtering, and rollover-budget search, LaVA achieves comparable generation fidelity to decode-based verification while providing up to a $10.7\times$ reduction in wall-clock time and a $17.5\times$ reduction in FLOPs. Reinvesting these savings into larger search populations further improves generation quality, reaching an OpenL3-FD of 64.62 on TangoFlux. These results demonstrate that efficient latent-space verification can make inference-time scaling substantially more practical for audio generation. Beyond computational savings, since the verifier operates on the latent state, its reward is differentiable with respect to the generator representation, potentially enabling future inference-time optimization without backpropagating through waveform decoding. More broadly, this approach could extend to other generative settings where expensive external evaluation limits inference-time search.

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